INCH-POUND

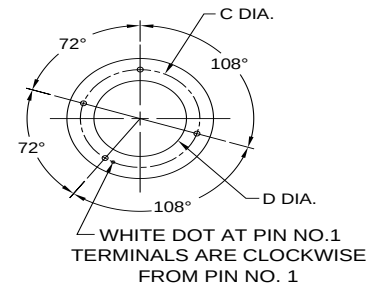
MIL-PRF-27/356E
AMENDMENT 1
21 May 2018
SUPERSEDING
MIL-PRF-27/356E
30 March 2015

PERFORMANCE SPECIFICATION SHEET

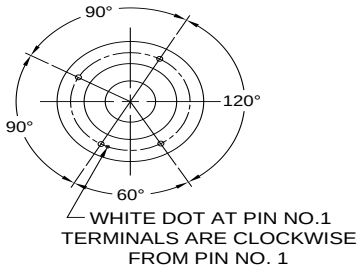
TRANSFORMERS AND INDUCTORS (AUDIO, POWER AND HIGH-POWER PULSE), INDUCTORS, POWER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

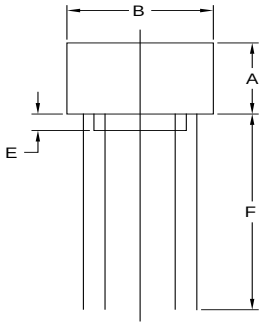
The complete requirements for procuring the transformer described herein shall consist of this document and the latest issue of specification MIL-PRF-27.

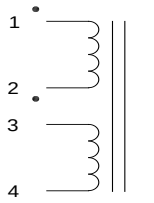


TERMINAL ARRANGEMENT A



TERMINAL ARRANGEMENT B





CIRCUIT DIAGRAM

FIGURE 1. Dimensions and configurations.

AMSC N/A

FSC 5950

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Dash number	A Max.	B Max. Dia.	C ±.010	D Max.	E	F	INSERT	Terminal Arrangement	Terminal diameter inches ±.002
01, 06, 11, 16, 21, 26, 31, 36, 41, 46, 51, 56, 61	.280 (7.11)	.410 (10.41)	.260 (6.6)	.250 (6.35)	.020 ±.010 (0.51±0.25)	1.00 ±0.28 (2.54 ± 6.35)		A	.012
02, 07, 12, 17, 22, 27, 32, 37, 42, 47, 52, 57, 62	.340 (8.63)	.500 (12.7)	.350 (8.89)	.340 (8.64)	.020 ±.010 (0.51 ±0.25)	1.00 ±0.28 (2.54 ± 6.35)		A	.012
03, 08, 13, 18, 23, 28, 33, 38, 43, 48, 53, 58, 63	.415 (10.54)	.630 (16.0)	.440 (11.17)	.430 (10.92)	.020 ±.010 (0.51 ±0.25)	1.00 ±0.28 (2.54 ± 6.35)		A	.016
04, 09, 14, 19, 24, 29, 34, 39, 44, 49, 54, 59	.500 (12.70)	.800 (20.32)	.560 (14.22)	.550 (13.97)	.020 ±.010 (0.51 ±0.25)	1.00 ±0.28 (2.54 ± 6.35)		B	.016
05, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60	.635 (16.12)	.950 (24.13)	.670 (17.0)	.660 (16.76)	.020 ±.010 (0.51 ±0.25)	1.00 ±0.28 (2.54 ± 6.35)	.138-32 UNC-2B .200 Min deep	B	.020

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking shall be on the side or on top.
4. For series connection join terminals 2 and 3, for parallel connection join terminals 1 and 3, 2 and 4.
5. Metric equivalents are in parentheses.

FIGURE 1. Dimensions and configurations - Continued.

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TABLE I. Electrical ratings.

Dash number	1/ Series inductance (mH)	Series dc current (max) (amps)	Series dc resistance ±25% (ohms)	1/ Parallel inductance (mH)	Parallel dc current (max) (amps)	Parallel dc resistance ±25% (ohms)
-01	20	.06	54	5	.12	13.5
-02	20	.10	29	5	.20	7.25
-03	20	.14	20	5	.28	5.0
-04	20	.22	12.4	5	.44	3.1
-05	20	.29	10.2	5	.58	2.55
-06	15	.07	40.5	3.75	.14	10.125
-07	15	.12	22	3.75	.24	5.5
-08	15	.16	15	3.75	.32	3.75
-09	15	.25	9.3	3.75	.50	2.325
-10	15	.33	7.7	3.75	.66	1.925
-11	10	.08	27	2.5	.16	6.75
-12	10	.14	14.6	2.5	.28	3.65
-13	10	.20	10	2.5	.40	2.5
-14	10	.31	6.2	2.5	.62	1.55
-15	10	.40	5.1	2.5	.80	1.275
-16	7.5	.10	20.3	1.88	.20	5.075
-17	7.5	.17	11	1.88	.34	2.75
-18	7.5	.24	7.5	1.88	.48	1.875
-19	7.5	.38	4.7	1.88	.76	1.175
-20	7.5	.49	3.8	1.88	.98	0.95
-21	5.0	.12	13.5	1.25	.24	3.375
-22	5.0	.21	7.3	1.25	.42	1.825
-23	5.0	.30	5	1.25	.60	1.25
-24	5.0	.47	3.1	1.25	.94	.775
-25	5.0	.6	2.6	1.25	1.2	.65
-26	3.0	.16	8.1	.75	.32	2.025
-27	3.0	.27	4.4	.75	.54	1.1
-28	3.0	.38	3	.75	.76	.75
-29	3.0	.6	1.9	.75	1.2	.475
-30	3.0	.78	1.5	.75	1.56	.375
-31	2.0	.19	5.4	.5	.38	1.35
-32	2.0	.33	2.9	.5	.66	.725
-33	2.0	.47	2	.5	.94	.5
-34	2.0	.74	1.3	.5	1.48	.325
-35	2.0	.95	1	.5	1.9	.25

See footnote at end of table.

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TABLE I. Electrical ratings – Continued.

Dash number	1/ Series inductance (mH)	Series dc current (max) (amps)	Series dc resistance ±25% (ohms)	1/ Parallel inductance (mH)	Parallel dc current (max) (amps)	Parallel dc resistance ±25% (ohms)
-36	1	.27	2.7	.25	.54	.675
-37	1	.47	1.5	.25	.94	.375
-38	1	.66	1	.25	1.32	.25
-39	1	1	.62	.25	2	.155
-40	1	1.3	.5	.25	2.6	.125
-41	.750	.31	2	.1875	.62	.5
-42	.750	.54	1	.1875	1.08	.25
-43	.750	.76	.75	.1875	1.52	.1875
-44	.750	1.2	.47	.1875	2.4	.1175
-45	.750	1.6	.38	.1875	3.2	.095
-46	.500	.38	1.3	.125	.76	.325
-47	.500	.66	.66	.125	1.32	.165
-48	.500	.93	.5	.125	1.86	.125
-49	.500	1.47	.31	.125	2.94	.0775
-50	.500	1.9	.25	.125	3.8	.0625
-51	.250	.53	.66	.0625	1.06	.165
-52	.250	.93	.33	.0625	1.86	.0825
-53	.250	1.3	.25	.0625	2.6	.0625
-54	.250	2.08	.16	.0625	4.16	.04
-55	.250	2.7	.13	.0625	5.4	.0325
-56	.100	.78	.30	.025	1.56	.075
-57	.100	1.35	.17	.025	2.7	.043
-58	.100	1.95	.11	.025	3.9	.028
-59	.100	3.08	.07	.025	6.16	.018
-60	.100	3.88	.06	.025	7.76	.015
-61	.040	1.18	.14	.01	2.36	.035
-62	.040	2.03	.07	.01	4.06	.018
-63	.040	2.68	.06	.01	5.36	.015

1/ The inductance must be at least 80 percent of the listed value when measured with rated dc current at 0.1 V rms, 10 kHz.

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REQUIREMENTS: (When numbers in parentheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings: See table I.

Design and construction:

Dimensions and configurations: See figure 1.

Duty cycle: Continuous.

Case: Epoxy.

Terminals: Printed circuit, solderable.

Weight: 1.25 ounces, maximum.

Altitude: 75,000 feet, maximum.

Operating temperature range: -55°C to +130°C.

Terminal strength: MIL-STD-202-211, test condition A, 2 pounds.

Dielectric withstanding voltage:

At sea level: 200 volts rms.

At reduced barometric pressure: 100 volts rms.

Vibration (high frequency): MIL-STD-202, method 204.

Magnetic shield: All units are magnetically shielded.

Temperature rise: 30°C with .1 volt rms, rated dc current as specified in table I at an ambient temperature of 100°C, maximum.

Marking location: See figure 1.

Part or Identifying Number (PIN): M27/356-(dash number from table I).

Qualification: Qualification testing and approval to M27/356-01 shall be sufficient to grant qualification approval to M27/356-01 through M27/356-63.

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Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-PRF-27, this document references the following:

MIL-STD-202-204 MIL-STD-202-211

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5950-2018-005)

Review activities:
Army – AR, MI
Navy - AS, MC, OS
Air Force - 19, 99

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